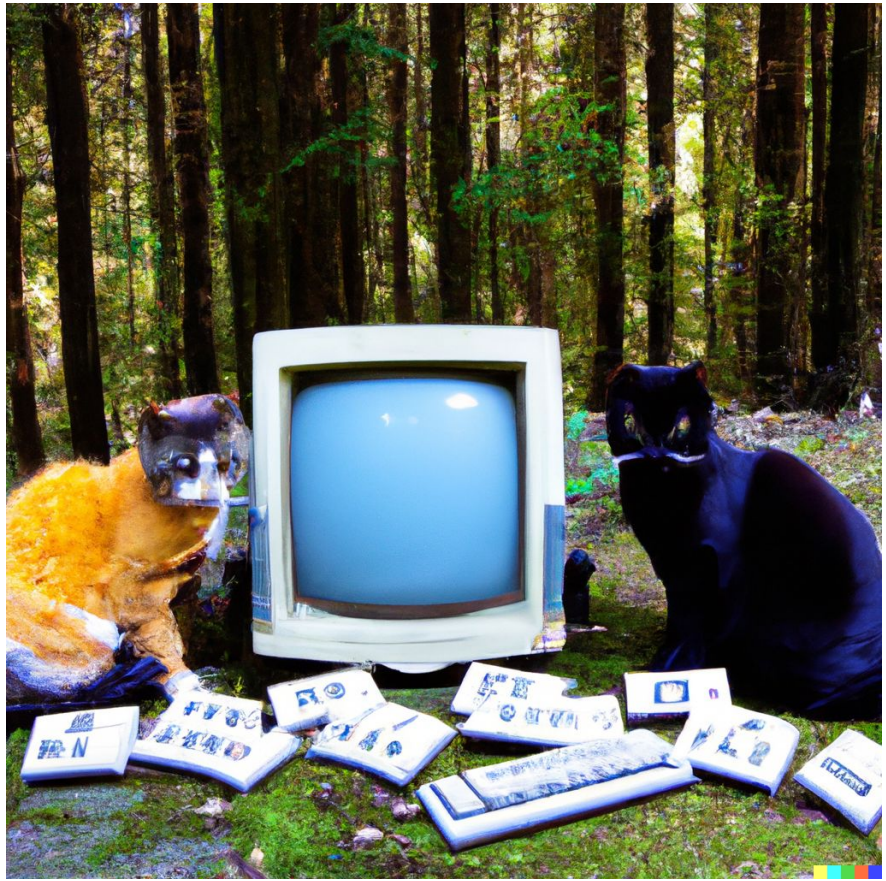
CS 103000
Prof. Madeline Blount

Week 6:
VECTORS



Dall-E 2: cats learning C++ in the forest on '90's technology



Programmers while reviewing the codes

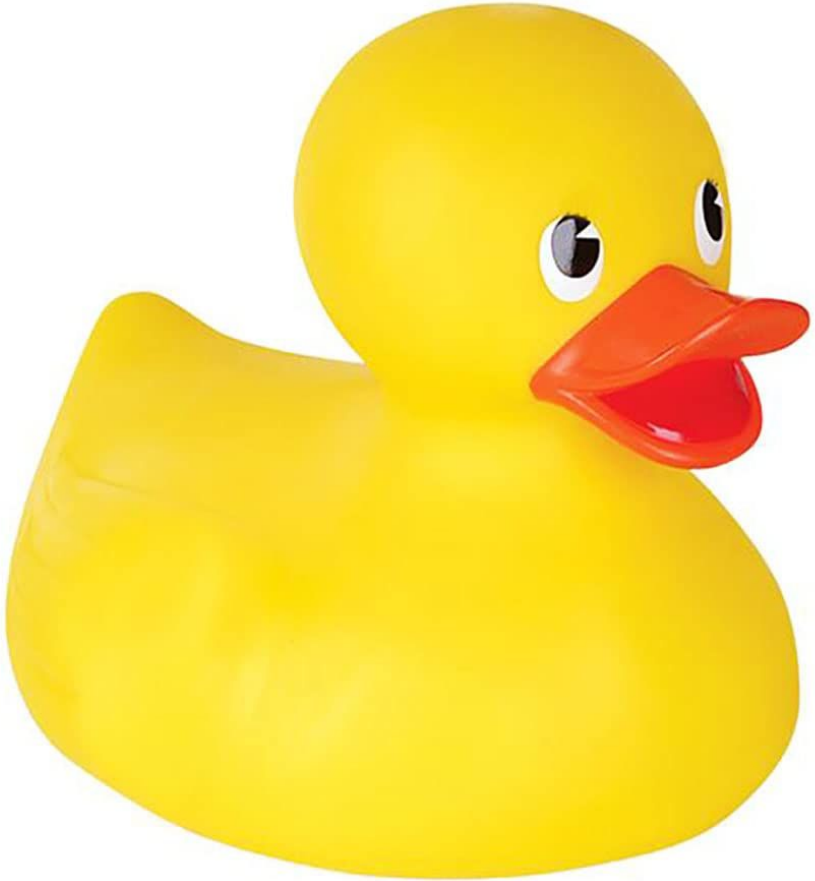


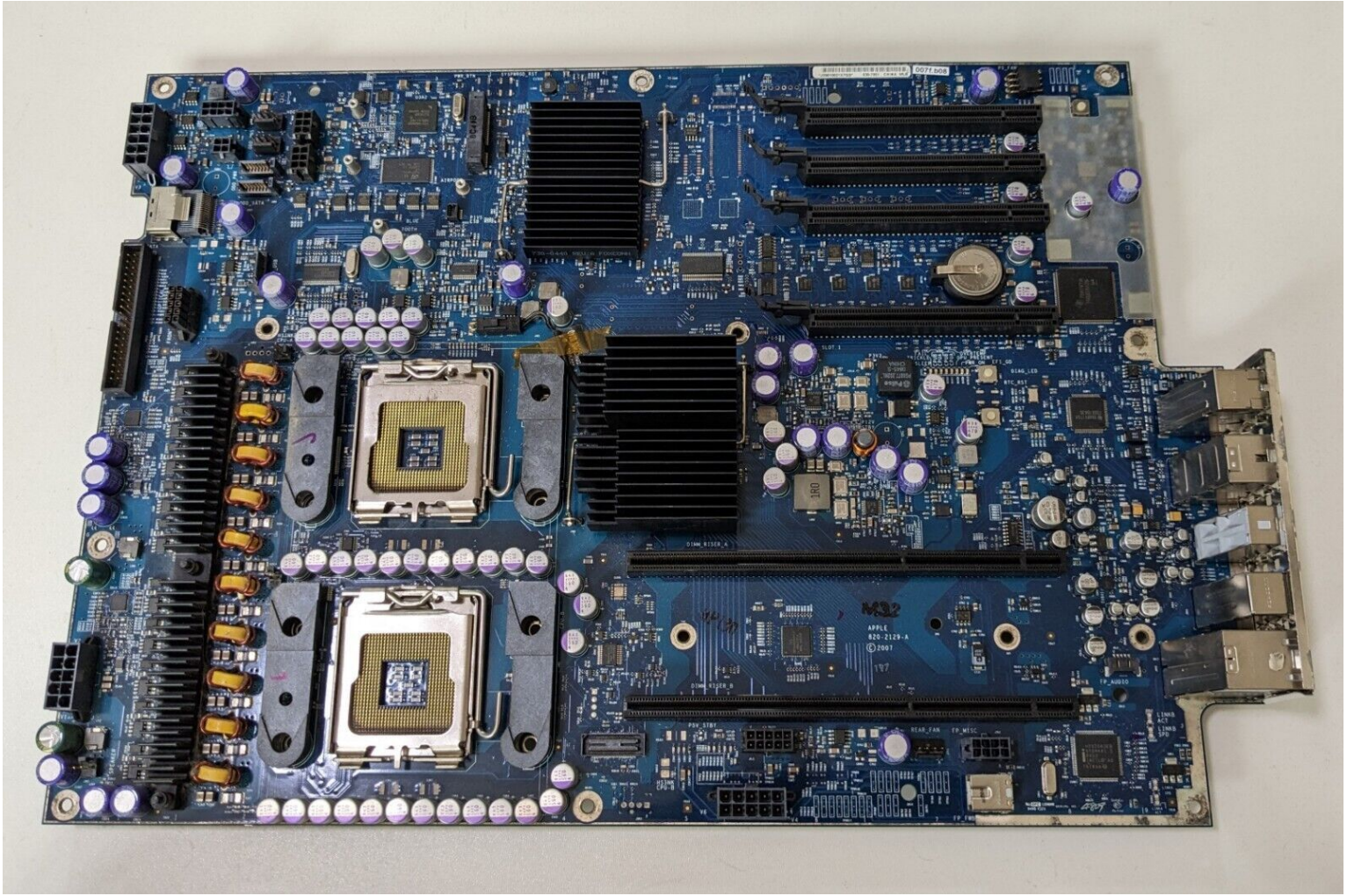

WHEN YOU REALIZE YOU'VE
BEEN STARING AT THE CODE
FOR HOURS
AND THE ERROR WAS
JUST A MISSING COMMA

When there's a bug in your code:

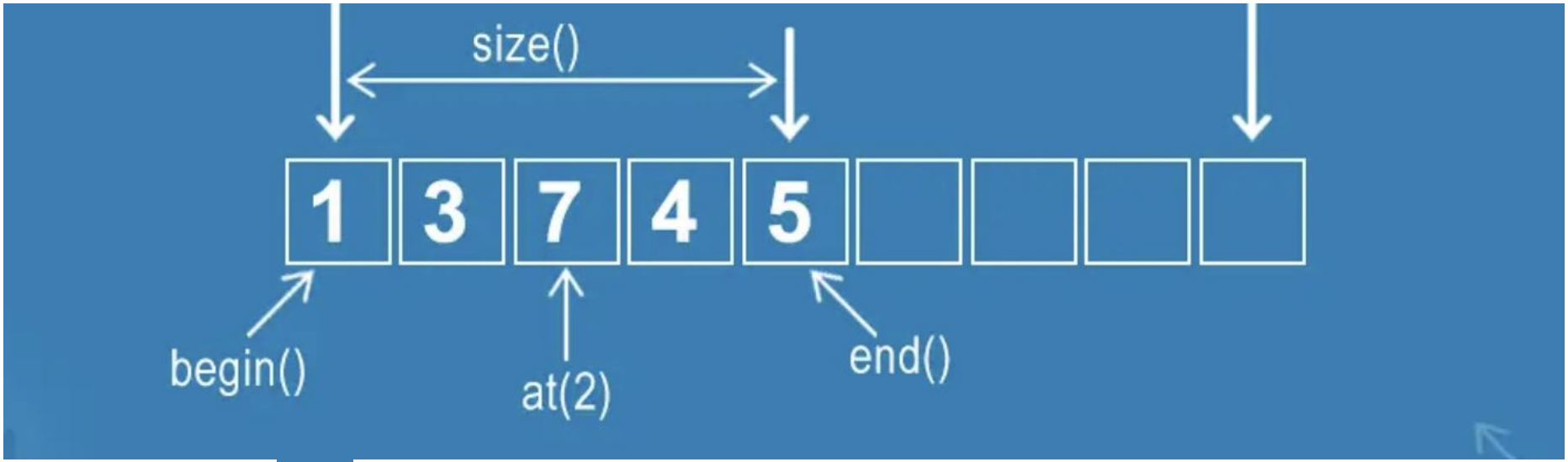







M32
APPLE
820-2129-A
©2007

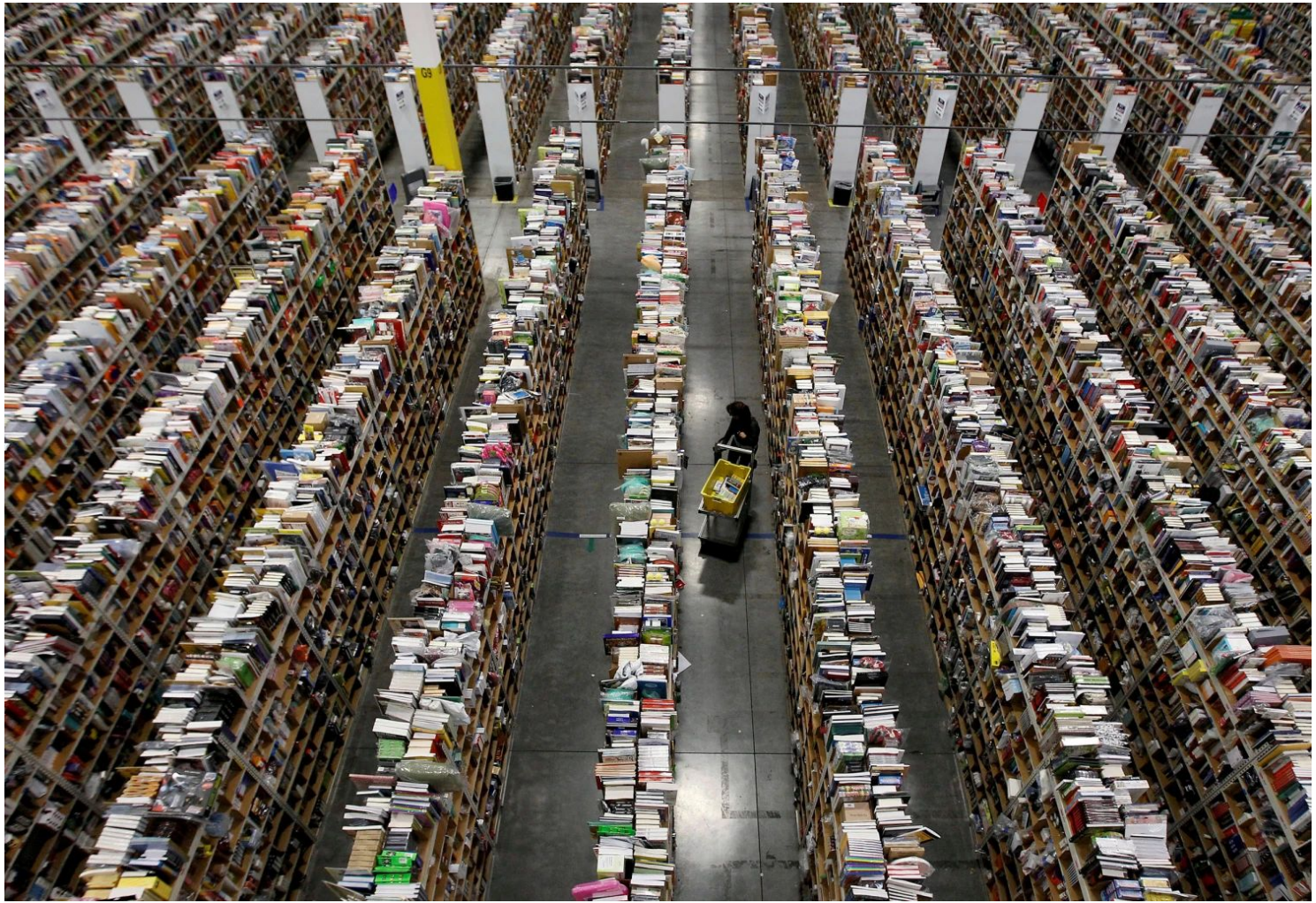
vectors!

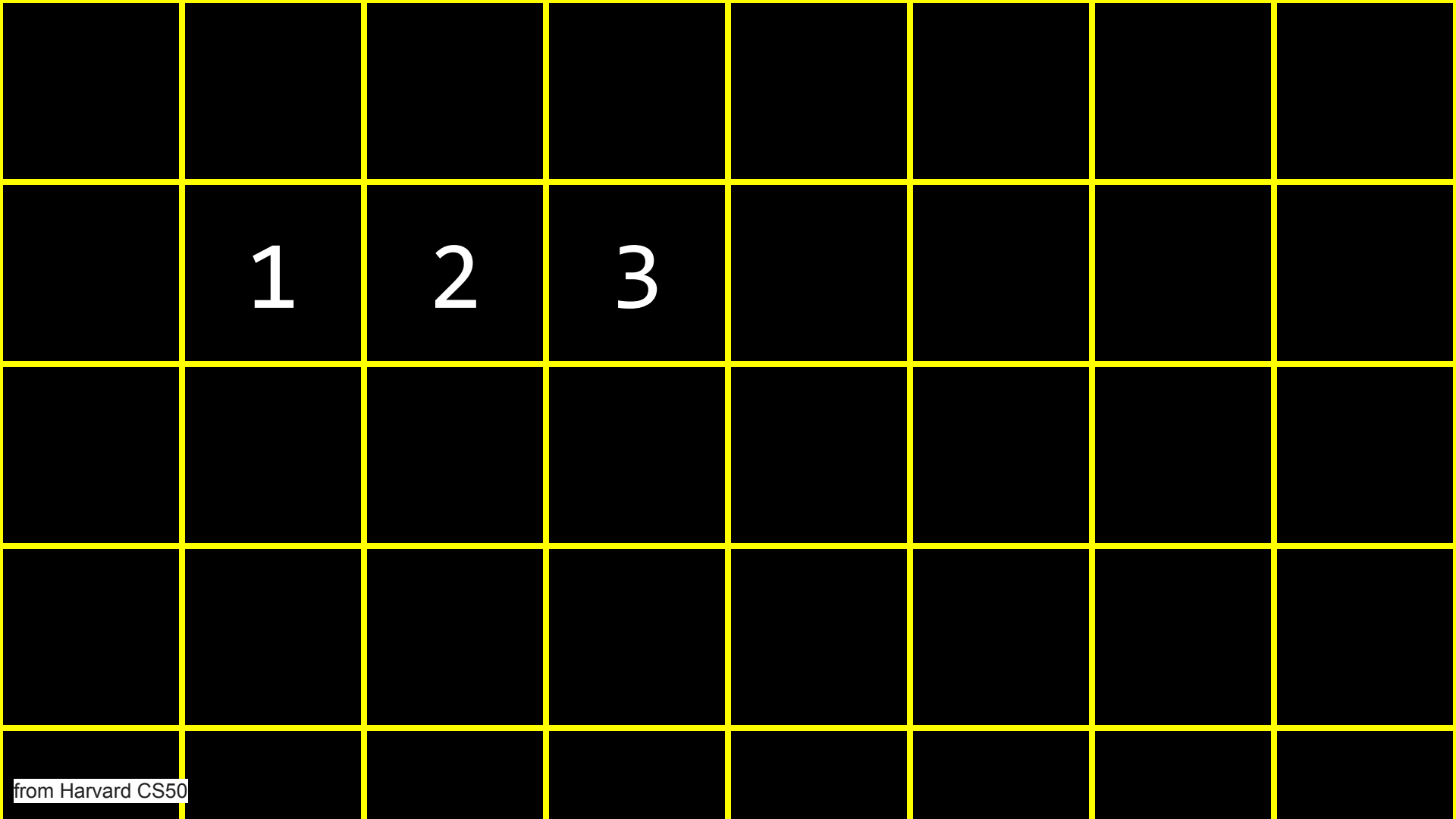



train.at(4)
train.at(2)
train.at(0)
COSCO
HYUNDAI
HYUNDAI
HYUNDAI

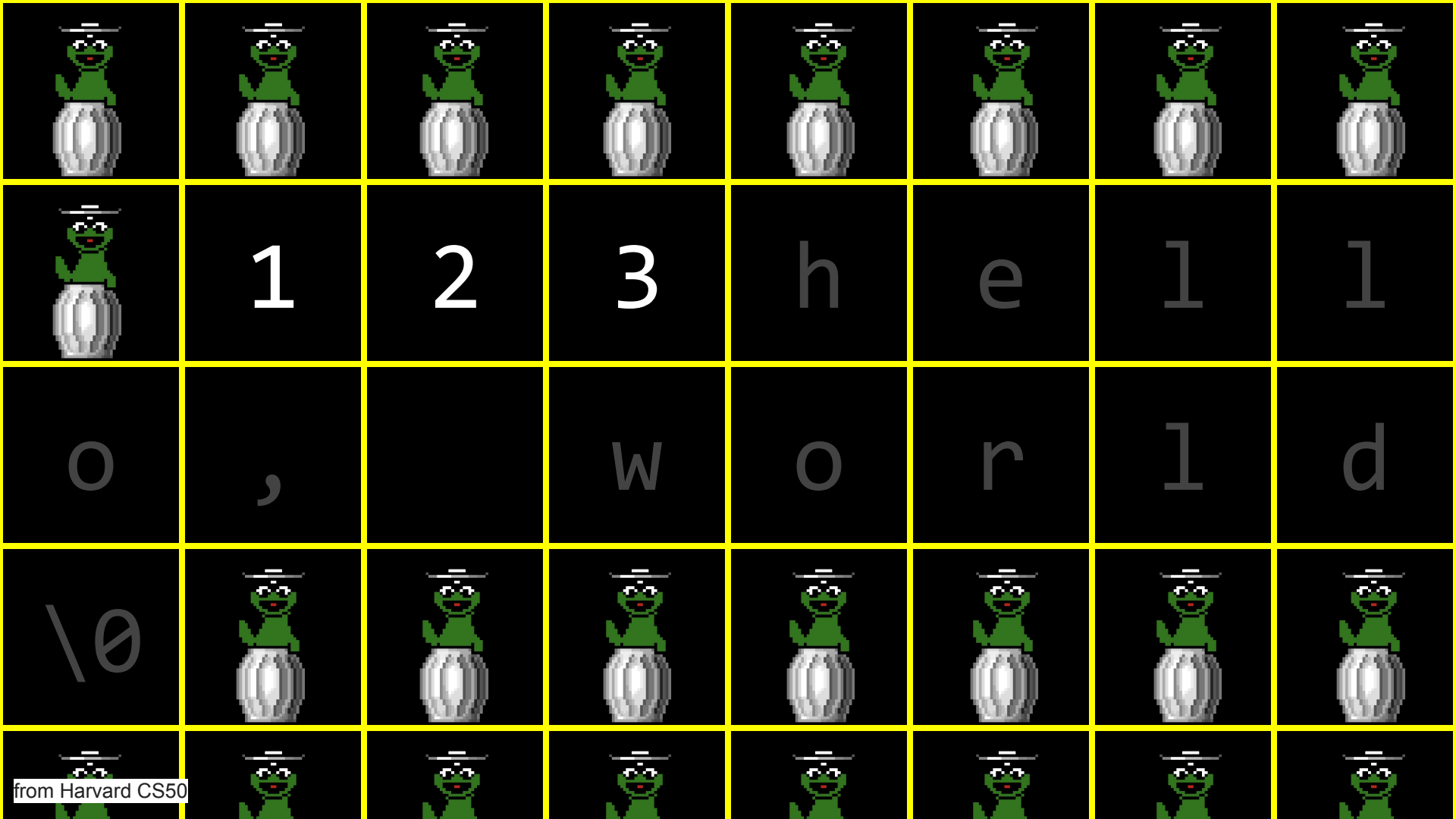

mail.at(0)
mail.at(1)
mail.at(2)
mail.at(3)
mail.at(4)
mail.at(5)
203
204
302
303
304


apt.at(0)
apt.at(2)





from Harvard CS50



from Harvard CS50

1	2	3



from Harvard CS50

1	2	3

1			

from Harvard CS50

1	2	3

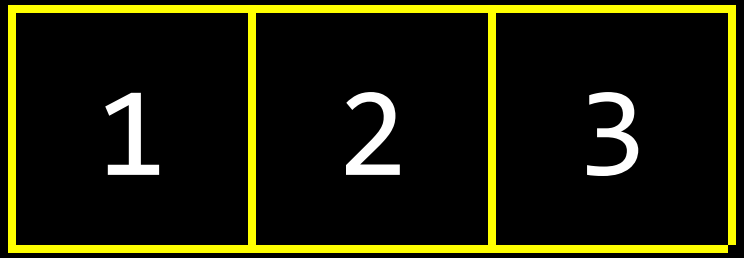
1	2		

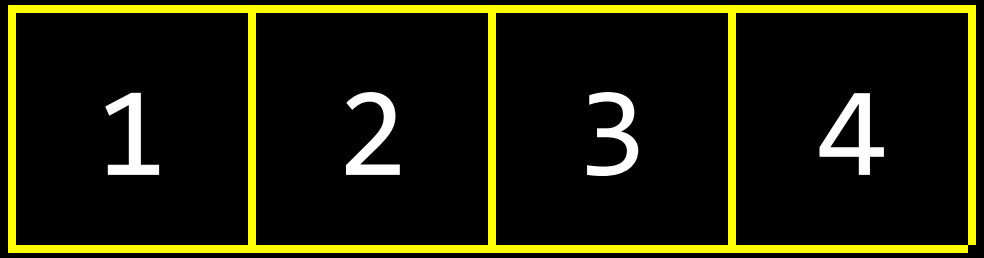
from Harvard CS50

1	2	3

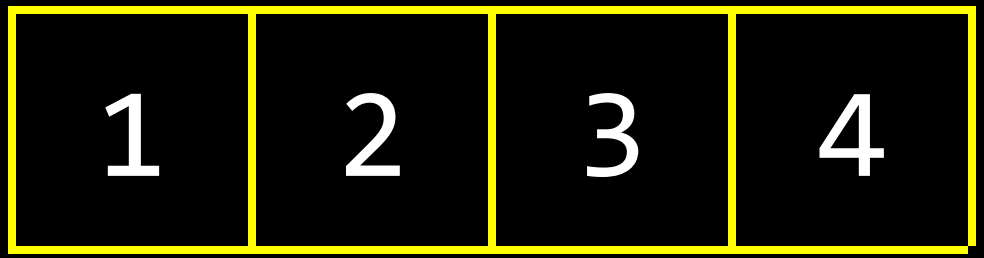
1	2	3	

from Harvard CS50





from Harvard CS50



from Harvard CS50

vectors vs. arrays?



```
vector<int> myNumbers(20);

int myNumbers[20];
```

vectors vs. arrays

- **BIG DIFFERENCE:**
  YOU CANNOT SIMPLY RESIZE ARRAYS!

- This makes arrays faster, if you are *really* in need of speedy performance (large, large datasets)

- Arrays don’t need a header #include

- For our purposes, simpler to use dynamic vectors

myContainer.at(i) vs. myContainer[i]

- .at() function checks the size of your container
- [] does not check the range!
- Both work for vectors
- Only [] works for built-in arrays …

C-strings vs. **strings** (C++)

```
char myWord[6] = “hello”;
string myWord = “hello”;
```

- C-string = older, from C, built-in
- Literally array of characters, with ‘\0’ to **END** (size = +1)
- Different library of functions than C++ strings
- Easy to make mistakes with!

Why learn arrays + C-strings?

- We still see them in code (“legacy”)
- Good to understand the most basic data types (like … binary) to know where our more advanced features come from!
- Vectors came from limitations with arrays, the standard template string from limitations with C-strings, etc.!



https://airtable.com/appLDEvuelyDRoI2b/shrxZpqfbjZin79fC

HACKATHON #1!

- Oct. 16: review session in lab, suggestions of what to study
- Oct. 19: release the prompt + groups! work in class
- Oct. 21: full work day in class
- Oct. 23: work day in class; 3 min. SHORT TALK by each group, work in progress; code due by end of night 11:59pm